

# Cryptography Exercises

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## 1 Statistical distance and information-theoretic encryption

Let  $A$  and  $B$  be two random variables taking values in a finite nonempty set  $\mathcal{X}$ . Define the *statistical distance*<sup>1</sup> between  $A$  and  $B$  as

$$\Delta(A; B) := \frac{1}{2} \cdot \sum_{a \in \mathcal{X}} |\Pr[A = a] - \Pr[B = a]|,$$

and *statistical distance of  $A$  from uniformity* as

$$d(A) := \Delta(A; U_{\mathcal{X}}),$$

where  $U_{\mathcal{X}}$  is uniform over  $\mathcal{X}$ . The reference uniform distribution is always on the declared set; for a tuple, it is on the Cartesian product of the declared sets for its coordinates. Analogously,  $\Delta$  can be defined as a distance between two *probability distributions*  $\alpha, \beta : \mathcal{X} \rightarrow [0, 1]$  as

$$\Delta(\alpha; \beta) := \Delta(A; B),$$

where  $A$  and  $B$  are random variables (taking values from  $\mathcal{X}$ ) distributed according to  $\alpha$  and  $\beta$  (respectively). A similar convention applies to the distance  $d$  from uniformity.

Randomized machines use private randomness independent of all external random choices.

### Exercise 1.1 An alternative definition of statistical distance

Show that for every  $A$  and  $B$  we have

$$\Delta(A; B) = \sum_{a \in \mathcal{X} : \Pr[A=a] > \Pr[B=a]} (\Pr[A = a] - \Pr[B = a]).$$

### Exercise 1.2 Statistical distance as a metric

Let  $\Pi$  be the set of all probability distributions over a finite nonempty set  $\mathcal{X}$ . Prove that  $\Delta$  is a *metric* on this set, i.e., it satisfies the following axioms:

- *non-negativity*: for every  $\alpha, \beta \in \Pi$  we have  $\Delta(\alpha; \beta) \geq 0$ ,
- *identity of indiscernibles*: for every  $\alpha, \beta \in \Pi$  we have that  $\Delta(\alpha; \beta) = 0$  if and only if  $\alpha = \beta$ ,
- *symmetry*: for every  $\alpha, \beta \in \Pi$  we have that  $\Delta(\alpha; \beta) = \Delta(\beta; \alpha)$ , and
- *triangle inequality*: for every  $\alpha, \beta, \gamma \in \Pi$  we have that  $\Delta(\alpha; \gamma) \leq \Delta(\alpha; \beta) + \Delta(\beta; \gamma)$ .

### Exercise 1.3 Statistical distance as distinguishing advantage

Let  $A$  and  $B$  be two random variables taking values from some finite set  $\mathcal{X}$ . For a (possibly randomized) machine  $\mathcal{D}$  that takes inputs from  $\mathcal{X}$  and outputs a bit, define its *distinguishing advantage* between  $A$  and  $B$  as

$$\text{Adv}_{\mathcal{D}}^{\text{dist}}(A, B) := |\Pr[\mathcal{D}(A) = 1] - \Pr[\mathcal{D}(B) = 1]|.$$

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<sup>1</sup> This is also known as the *total variation distance*, especially in the quantum information literature

Prove that for every computationally unbounded machine  $\mathcal{D}$  we have

$$\text{Adv}_{\mathcal{D}}^{\text{dist}}(A, B) \leq \Delta(A; B). \quad (1)$$

For every  $A$  and  $B$ , construct  $\mathcal{D}$  that achieves equality in Eq. (1). Show that the maximum is unchanged if the absolute value is omitted, and hence

$$\Delta(A; B) = \max_{\mathcal{D}} (\Pr[\mathcal{D}(A) = 1] - \Pr[\mathcal{D}(B) = 1]),$$

where the maximum is over all computationally unbounded machines that output a bit.

#### Exercise 1.4 Statistical distance and guessing advantage

Let  $\alpha_0$  and  $\alpha_1$  be two distributions over a finite nonempty set  $\mathcal{X}$ . Consider the following game played by a computationally unbounded, possibly randomized machine  $\mathcal{M}$  (that knows  $\alpha_0$  and  $\alpha_1$ ):

1. A uniformly random bit  $b \leftarrow \{0, 1\}$  is chosen.
2. A value  $A$  is sampled according to distribution  $\alpha_b$  and sent to  $\mathcal{M}$ .
3.  $\mathcal{M}$  receives  $A$  and produces output  $\hat{b} \in \{0, 1\}$ .

We say that  $\mathcal{M}$  *won the game* if  $\hat{b} = b$ . Define its *guessing advantage* as

$$\text{Adv}_{\mathcal{M}}^{\text{guess}} := |\Pr[\mathcal{M} \text{ wins the game}] - \frac{1}{2}|.$$

Show that

$$\forall_{\mathcal{M}} \Pr[\mathcal{M} \text{ wins the game}] \leq \frac{1 + \Delta(\alpha_0; \alpha_1)}{2}. \quad (2)$$

For every  $\alpha_0$  and  $\alpha_1$ , construct  $\mathcal{M}$  that achieves equality in Eq. (2). By also considering the machine that complements its output, deduce that

$$\max_{\mathcal{M}} \text{Adv}_{\mathcal{M}}^{\text{guess}} = \frac{1}{2} \Delta(\alpha_0; \alpha_1).$$

#### Exercise 1.5 How function application changes statistical distance

Let  $A$  and  $B$  take values in a finite nonempty set  $\mathcal{X}$ , and let  $\mathcal{Y}$  be a finite nonempty set. Show that for every function  $f : \mathcal{X} \rightarrow \mathcal{Y}$  we have

$$\Delta(f(A); f(B)) \leq \Delta(A; B).$$

Show that if  $f$  is an injection, then

$$\Delta(f(A); f(B)) = \Delta(A; B).$$

Deduce that if  $f$  is a bijection from  $\mathcal{X}$  onto  $\mathcal{Y}$ , then  $d(f(A)) = d(A)$ .

#### Exercise 1.6 One-time pad with imperfect randomness

Let  $(\text{Enc}, \text{Dec})$  be the one-time pad encryption scheme for messages from  $\{0, 1\}^t$ . Suppose a key  $K$  is chosen from  $\{0, 1\}^t$  according to some distribution  $\alpha$ . Consider the following indistinguishability game played between a computationally unbounded adversary  $\mathcal{A}$  and a challenger  $\Omega$ :

1.  $\mathcal{A}$  produces two messages  $m_0, m_1 \in \{0, 1\}^t$  and sends them to  $\Omega$ .
2.  $\Omega$  selects  $b \leftarrow \{0, 1\}$  uniformly at random, independently samples a fresh key  $K$  according to  $\alpha$ , computes  $c = \text{Enc}(K, m_b)$ , and sends  $c$  to  $\mathcal{A}$ .
3.  $\mathcal{A}$  receives  $c$  and produces as output  $\hat{b} \in \{0, 1\}$ .

We assume that  $\mathcal{A}$  knows  $\alpha$ . We say that  $\mathcal{A}$  *won the game* if  $\hat{b} = b$ . Show that

$$\forall_{\mathcal{A}} \Pr[\mathcal{A} \text{ wins the game}] \leq \frac{1}{2} + d(\alpha).$$

**Exercise 1.7 Conditional statistical distance**

Let  $A$  and  $B$  take values in finite nonempty sets  $\mathcal{X}$  and  $\mathcal{Y}$ , respectively. For each  $b$  with  $\Pr[B = b] > 0$ , let  $\alpha_b$  be the conditional distribution of  $A$  given  $B = b$ , viewed as a distribution over  $\mathcal{X}$ . Define the *statistical distance of  $A$  from uniformity conditioned on  $B$*  as

$$d(A \mid B) := \sum_{b \in \mathcal{Y}: \Pr[B=b]>0} \Pr[B = b] \cdot d(\alpha_b).$$

Show that  $d(A \mid B) = \Delta((A, B); (U_{\mathcal{X}}, B))$ , where  $U_{\mathcal{X}}$  is uniform over  $\mathcal{X}$  and independent of  $B$ . Can you find a game-based interpretation of  $d(A \mid B)$  similar to the one in Exercise 1.4?

**Exercise 1.8 Conditioning does not decrease the distance from uniformity**

Show that for every two random variables  $A$  and  $B$  taking values in finite nonempty sets, we have  $d(A \mid B) \geq d(A)$ .